

Why Observational AI-Agent Pull Request Datasets Cannot Support Behavioral Inference: A Critical Analysis of Methodology Failures in Software Engineering Research

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Abstract—We examine the suitability of observational pull request datasets for AI agent behavioral research using the AIDev dataset as a case study. Our analysis reveals fundamental validity failures that preclude meaningful inference: agent attribution exhibits obvious errors (GitHub Copilot: 379 users vs. millions in reality), statistical tests violate independence assumptions due to clustering, and measurement approaches show systematic bias. We demonstrate why large-scale observational studies of AI tools cannot support behavioral claims without controlled experimental designs. This methodology critique provides guidelines for valid AI tool evaluation and warns against common pitfalls in software engineering research that mistake dataset size for validity.

Index Terms—methodology, software engineering research, AI tools, empirical evaluation, validity threats

I. INTRODUCTION

The software engineering research community increasingly relies on large observational datasets to study AI coding tools. The AIDev dataset exemplifies this trend, containing 932,791 pull requests with agent attribution labels that researchers have used to draw behavioral conclusions about different AI tools. However, such observational approaches face fundamental methodological challenges that compromise validity.

This paper examines whether pull request datasets can support reliable behavioral inferences about AI tools. Through systematic analysis of the AIDev dataset, we identify critical failures in: (1) agent attribution reliability, (2) statistical assumption compliance, and (3) measurement validity. These issues collectively demonstrate why observational PR data cannot support behavioral claims about AI tools.

A. The Appeal of Large Datasets

Large observational datasets offer compelling advantages: massive sample sizes, real-world authenticity, and comprehensive coverage. The AIDev dataset’s 932,791 observations across 72,189 developers appears to provide unprecedented statistical power for understanding AI tool behavior. This scale creates an illusion of methodological rigor that can mask fundamental validity threats.

B. Research Questions

We address three methodological research questions:

RQ1: What validity threats prevent reliable inference about AI agent behavior from observational PR datasets? We systematically identify attribution failures, statistical violations, and measurement biases.

RQ2: How do these validity threats manifest in practice using the AIDev dataset? We provide concrete examples of each threat category with specific evidence.

RQ3: What methodological requirements are necessary for valid AI tool behavioral research? We specify design principles for reliable causal inference.

II. METHODOLOGY FAILURES IN OBSERVATIONAL AI RESEARCH

A. RQ1: Systematic Validity Threats

1) *Agent Attribution Unreliability*: Observational datasets typically derive agent labels through metadata extraction, commit message parsing, and self-reports. This approach suffers from multiple failure modes:

Detection Incompleteness. Automated attribution can only identify explicitly declared tool usage, missing silent assistance or tools used without obvious signatures.

Temporal Inconsistency. Tool adoption changes rapidly, but retrospective labeling may use contemporary usage patterns to infer historical tool use.

Mixed Authorship. Real development involves human editing of AI-generated code, making pure agent attribution impossible in observational settings.

Self-Report Bias. Developers may over-report or under-report tool usage based on perceived social desirability or organizational policies.

2) *Statistical Independence Violations*: Standard statistical tests assume independent observations, but PR datasets exhibit multiple clustering structures:

Repository Clustering. PRs within repositories share coding standards, review processes, and maintainer preferences. These systematic similarities violate independence.